



YASH RANA

Roll No.: 25MTCS017

M.Tech. - Computer Science and Engineering (Data Science)

Indian Institute of Information Technology Una

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Github

LinkedIn

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech. (Data Science)	Indian Institute of Information Technology Una	Pursuing	2025 – 2027 (Expected)
B.Tech. CSE (Spl. AIML)	Chandigarh University, Punjab	7.21 / 10.0	2020 – 2024
Senior Secondary (10+2)	Bharti Vidyapeeth School Baijnath	83.20%	2020
Secondary (10th)	Bharti Vidyapeeth School Baijnath	90.80%	2018

EXPERIENCE

- Viper Acoustics** Jun. 2021 – Jan. 2022
Security Researcher (Collaborative Project) Audio Driver Software Security
 - Performed in-depth penetration testing and static/dynamic analysis on Viper4Android FX audio driver application, mitigating critical vulnerabilities and enhancing application security posture by 30%.
 - Enhanced signature verification mechanisms and evaluated root-access implications to prevent unauthorized bytecode tampering, resource modification, and remote arbitrary command execution.
 - Partnered with the development team to integrate secure SDLC best practices throughout the development lifecycle and audited integrated third-party libraries.

PROJECTS

- App-Patcher Framework** Python, Smali, APKEditor, Uber-Signer
Android Reverse Engineering Pipeline Github
 - Engineered an automated framework to decompile, patch, and repack split bundles (APKS/XAPK/APKM) into standalone signed APKs with multi-DEX smali bytecode manipulation and manifest XML injection.
- AnyLM** FastAPI, PyTorch, FAISS, LangChain
Private On-Device RAG Assistant Github
 - Architected an on-device RAG system supporting multi-source document ingestion, FAISS vector retrieval, local quantized LLM inference (TinyLlama), and SSE streaming generation with evaluation metrics.
- MountDroid** Shell, Linux USB Gadget, Android Kernel
Bootable ISO Virtualization via USB Gadget Github
 - Built a root utility utilizing Linux USB Mass Storage Gadget to expose ISOs over USB, enabling direct PC booting from phone storage across Android 15 AOSP and Nothing OS.
- PowerRate & Windows Utilities** Batch, Windows API, System Automation
Display Refresh Rate Automation Github
 - Built an automated Windows system utility that dynamically switches laptop display refresh rates between AC and battery power states to optimize battery endurance without manual intervention.

TECHNICAL SKILLS

- Programming & Scripting:** Python, C, C++, Java, Smali, Bash/Shell, PowerShell, SQL, JavaScript
- Security & Reverse Eng.:** IDA Pro, APK Decompilation/Patching, Static/Dynamic Analysis, Penetration Testing, Linux Gadget
- AI/ML & Frameworks:** PyTorch, Hugging Face Transformers, LangChain, FAISS, FastAPI, Splunk, Linux/Android Internals, Git

ACHIEVEMENTS

- Instagram Web Security:** Identified and investigated a security vulnerability in Instagram web platform. 2024
- Windows Security Research:** Discovered multiple Windows security vulnerabilities and exploitable behaviors. 2024
- Open-Source Contributor:** Authored public tools for Android reverse engineering and USB virtualization. 2023–Present